

Generalized Unbiased Reconstruction for Gradient-Domain Rendering

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Fig. 1. We propose a generalized unbiased reconstruction method for gradient-domain rendering. We formulate a generalized reconstruction model based on linear combinations of full-image color and gradient information, under an unbiasedness condition to ensure theoretical unbiasedness of the reconstruction. A practical algorithm is then developed, featuring a truly unbiased implementation using a dual-buffer strategy and a slightly biased variant using a single buffer. Our method outperforms both the standard unbiased baseline (L_2 Poisson) and the slightly biased IDUW [Back et al. 2023] in terms of visual quality and accuracy.

Gradient-domain rendering estimates image-space gradients using correlated sampling, which can be combined with color information to reconstruct smoother and less noisy images. While simple L_2 reconstruction is unbiased, it often leads to visible artifacts. In contrast, most recent reconstruction

methods based on learned or handcrafted techniques improve visual quality but introduce bias, leaving the development of practically unbiased reconstruction approaches relatively underexplored.

In this work, we propose a generalized framework for unbiased reconstruction in gradient-domain rendering. We first derive the unbiasedness condition under a general formulation that linearly combines pixel colors and gradients. Based on this unbiasedness condition, we design a practical algorithm¹ that minimizes image variance while strictly satisfying unbiasedness. Experimental results demonstrate that our method not only guarantees unbiasedness but also achieves superior quality compared to existing unbiased and slightly biased reconstruction methods.

CCS Concepts: • Computing methodologies → Rendering; Ray tracing.

Additional Key Words and Phrases: Gradient-Domain Rendering, Reconstruction, Optimization

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¹Our method is open-sourced at <https://github.com/lastmc/GURecon>.

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1 Introduction

Gradient-domain rendering [Lehtinen et al. 2013; Kettunen et al. 2015], a powerful extension of photorealistic rendering, estimates not only pixel colors but also their spatial gradients by tracing correlated light paths. These estimates can be further combined to reconstruct the final image by solving a screened Poisson equation [Pérez et al. 2023]. This reconstruction paradigm can significantly reduce variance and improve visual quality.

A central requirement in Monte Carlo rendering is unbiasedness—ensuring that the expected reconstructed image equals the true solution of the rendering equation. However, most existing reconstruction and denoising methods for gradient-domain rendering violate this principle. While recent methods employ sophisticated learned or handcrafted techniques to reduce noise, they often introduce bias [Kettunen et al. 2019; Guo et al. 2019; Yan et al. 2024] or achieve only slight biasedness [Back et al. 2023]. Currently, the only reconstruction method that guarantees strict unbiasedness is the $\mathcal{L}_2$ screened Poisson reconstruction, which minimizes an energy function involving both pixel color and gradient estimates. Unfortunately, this method suffers from slow convergence and visible artifacts, particularly near high-frequency edges, greatly limiting its practical use.

In this work, we address this limitation by providing both a theoretical foundation and a practical algorithm for *unbiased* image reconstruction. Theoretically, we introduce a generalized framework for linear reconstruction from Monte Carlo color and gradient estimates. We prove that a pixel’s color can be unbiasedly estimated through a linear combination of all pixel colors and gradients, provided the combination weights satisfy a specific linear constraint—which we refer to as the *unbiasedness condition*. This formulation significantly enlarges the solution space for unbiased reconstructions. Notably, we demonstrate that the $\mathcal{L}_2$ reconstruction is one of the feasible solutions in this solution space, also satisfying our generalized unbiasedness condition.

Practically, we propose a reconstruction method that minimizes output variance subject to our unbiasedness condition. This results in a quadratic optimization problem with linear constraints, which we solve efficiently as a sparse linear system. To ensure true unbiasedness during numerical computation, we develop a dual-buffer implementation strategy. Our method delivers high-quality image reconstruction results and consistently outperforms existing unbiased methods in terms of visual fidelity and noise suppression.

Our contributions are:

- A general formulation of linear reconstruction in Monte Carlo rendering (§4.1).
- A necessary and sufficient condition for unbiased reconstruction (§4.2 and §4.3).

- A practical, variance-minimizing reconstruction algorithm designed for gradient-domain rendering, which satisfies the unbiasedness condition and achieves state-of-the-art performance (§5).

2 Related Work

Our work focuses on unbiased image reconstruction for gradient-domain rendering. We discuss the relevant prior work of these topics in this section. For comprehensive reviews, readers may refer to surveys on gradient-domain rendering by Hua et al. [2019], and on Monte Carlo denoising by Zwicker et al. [2015] and Yen et al. [2025].

2.1 Gradient-Domain Rendering

Image-space gradients — differences between neighboring pixels — encode first-order structural image information. They help delineate sharp edges and segment smooth regions and have long been used in image processing [Kou et al. 2015; Bhat et al. 2010; Kazhdan and Hoppe 2008]. In physically based rendering, Lehtinen et al. [2013] first introduced gradient-domain techniques to address uneven convergence in Metropolis light transport [Veach and Guibas 1997]. Image gradients, estimated using pairs of correlated light paths, are then used with pixel colors to reconstruct the final image. The most common reconstruction strategy involves solving a screened Poisson equation [Pérez et al. 2023]. This framework was subsequently extended to various light transport algorithms, including unidirectional path tracing [Kettunen et al. 2015; Bauszat et al. 2017], bidirectional path tracing [Manzi et al. 2015], photon mapping [Hua et al. 2017], and vertex connection and merging [Sun et al. 2017]. These methods all share a common framework: correlated sampling of gradients and colors, followed by an image-space reconstruction step.

For further improvement, Manzi et al. [2014] proposed an enhanced sampling strategy that avoids estimating gradients in regions where shift mappings become unreliable, thereby reducing noise and bias in challenging cases. Similar ideas were later explored in gradient-domain path reuse [Bauszat et al. 2017], and gradient-domain bidirectional path tracing [Manzi et al. 2015], all of which emphasize improving gradient estimators through carefully designed sampling schemes.

These approaches primarily aim to improve the efficiency of Monte Carlo sampling by designing more effective gradient estimators and correlation strategies. However, such improvements often rely on problem-specific heuristics and can be difficult to generalize across different rendering algorithms. In contrast, our work targets the reconstruction step, aiming to improve the robustness and quality of the final image regardless of the underlying sampling scheme.

The statistical correlation between pixel colors and gradients enables significant variance reduction. A notable example is the $\mathcal{L}_2$ screened Poisson reconstruction, which combines both estimates without introducing bias. However, despite being theoretically unbiased, $\mathcal{L}_2$ reconstruction often produces visible artifacts around high-frequency regions (e.g., edges and detailed structures) and suffers from slow convergence at low sample counts. In contrast, the $\mathcal{L}_1$ variant offers better perceptual quality but introduces bias.

More recent techniques have explored sophisticated reconstruction strategies. Methods such as NGPT [Kettunen et al. 2019], Grad-Net [Guo et al. 2019], DeepCombiner [Back et al. 2020], and FRGR [Yan et al. 2024] are either biased by design or achieve only asymptotic unbiasedness. The method by Back et al. [2023] is slightly biased, as it assumes sampled pixel colors follow symmetric distributions, which is rarely satisfied in practice and thus leads to bias. These biased reconstruction techniques motivate our work to develop a reconstruction method that is both provably unbiased and perceptually effective.

2.2 Monte Carlo Denoising

The noise inherent in Monte Carlo rendering typically requires either increasing sample counts or applying post-processing denoising techniques. Image-space denoising has gained popularity due to its generality and efficiency. These denoising methods operate entirely in the image space, independent of the underlying path sampling algorithm or scene complexity.

Early work mainly used manually designed filters. Using auxiliary buffers (e.g., albedo, normal, and depth), filter-based approaches [Kopf et al. 2007; Rousselle et al. 2013; Moon et al. 2014; Bitterli et al. 2016] estimate pixel similarity to compute weights of every pixel for smoothing or regression. The advent of deep learning brought significant improvements to Monte Carlo denoising [Kalantari et al. 2015; Bako et al. 2017; Vogels et al. 2018]. For example, KPCN [Bako et al. 2017] proposed learning filter kernels using convolutional neural networks (CNNs) conditioned on auxiliary buffers. While effective, such kernel-prediction-based methods are limited by the receptive fields of kernels; increasing the kernel sizes often incurs prohibitive computational costs. Subsequent work addressed this issue by directly predicting denoised images. For example, AFGSA [Yu et al. 2021] used self-attention to dynamically model long-range dependencies, while NPPD [Balint et al. 2023] employed hierarchical feature pyramids for real-time denoising.

Although these denoising approaches do not use gradient information directly, they are relevant in the broader context of reconstructing low-variance, high-fidelity (but biased) images from noisy samples. In particular, insights from their feature-based filtering and perceptual loss functions can potentially inform improved design for unbiased reconstruction methods.

3 Motivation and Overview

In physically based rendering, the pixel color I_i^* is formulated as a path integral [Veach 1997]:

$$I_i^* = \int f_i(\bar{\mathbf{x}}) d\mu(\bar{\mathbf{x}}), \quad (1)$$

where f_i is the measurement contribution function of the pixel i , $\bar{\mathbf{x}}$ denotes a light path, and μ is the area-product measure. The pixel color difference (gradient) between pixel i and pixel j is defined as $\Delta_{ij}^* := I_j^* - I_i^*$. To compute these integrals, a common practice is using Monte Carlo integration, resulting in unbiased estimates of pixel colors I_i and pixel differences Δ_{ij} such that

$$\mathbb{E}(I_i) = I_i^* \quad \text{and} \quad \mathbb{E}(\Delta_{ij}) = \Delta_{ij}^*. \quad (2)$$

Due to the limited number of Monte Carlo samples, the estimates I_i and Δ_{ij} are often noisy. The goal of image reconstruction is to generate a “denoised” image with higher visual quality by combining those noisy estimates. There is large space for designing reconstruction algorithms. In particular, we want to answer the following two questions.

- (1) **Unbiasedness:** Since an unbiased Monte Carlo estimator is often desired, how can we ensure that our reconstruction algorithm is also unbiased?
- (2) **Optimality:** To improve image quality, what is the optimal way to combine the noisy estimates of pixel colors and their differences?

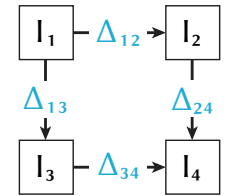
3.1 Overview of Our Method

In this work, we propose an unbiased, variance-minimizing reconstruction algorithm for gradient-domain rendering. In §4, we introduce the unbiasedness condition of generalized linear image reconstruction, which is expressed as a linear system of reconstruction weights. In §5, we formulate the reconstruction problem as a constrained optimization, and we show how to find the optimal reconstruction weights by minimizing the sample variance of Monte Carlo rendering, while satisfying the unbiasedness condition. Finally, we evaluate the design choices of our method and compare it with existing reconstruction techniques for gradient-domain rendering in §6.

Before jumping into technical details, we first present a toy example that reconstructs a 2×2 image to provide some intuition about our method.

3.2 Toy Example: Reconstructing a 2×2 Image

For the 2×2 image shown on the right, we define the pixel colors I_1, I_2, I_3, I_4 and the pixel gradients $\Delta_{12}, \Delta_{13}, \Delta_{24}, \Delta_{34}$ as random variables, whose variances are listed in Table 1. Suppose that we want to reconstruct the color of pixel 1, there are five possible estimators by finding the shortest path from every pixel to pixel 1:



$$\begin{aligned} H^{(1)} &= I_1, \quad H^{(2)} = I_2 - \Delta_{12}, \quad H^{(3)} = I_3 - \Delta_{13}, \\ H^{(4)} &= I_4 - \Delta_{24} - \Delta_{12}, \quad H^{(5)} = I_4 - \Delta_{34} - \Delta_{13}. \end{aligned} \quad (3)$$

All of these estimators are unbiased, i.e., $\mathbb{E}(H^{(i)}) = \mathbb{E}(I_1)$ for $i = 1, \dots, 5$. As a result, any affine combination of these estimators,

$$H = \sum_{i=1}^5 c_i H^{(i)}, \quad \text{such that} \quad \sum_{i=1}^5 c_i = 1, \quad (4)$$

is also unbiased, i.e., $\mathbb{E}(H) = \mathbb{E}(I_1)$. By plugging Eqn. (3) and $\Delta_{ij} = I_j - I_i$ into Eqn. (4) and reorganizing the terms, we can express the combined estimator (i.e., the reconstructed pixel color) as a linear

Table 1. The variance settings of the 2×2 example.

Random Variable	I_1	I_2	I_3	I_4	Δ_{12}	Δ_{13}	Δ_{24}	Δ_{34}
Variance	1	4	4	8	2	2	2	2

combination of the pixel colors and gradients:

$$H = \underbrace{c_1}_{\alpha_1} I_1 + \underbrace{c_2}_{\alpha_2} I_2 + \underbrace{c_3}_{\alpha_3} I_3 + \underbrace{(c_4 + c_5)}_{\alpha_4} I_4 \\ + \underbrace{(-c_2 - c_4)}_{\beta_{12}} \Delta_{12} + \underbrace{(-c_3 - c_5)}_{\beta_{13}} \Delta_{13} + \underbrace{(-c_4)}_{\beta_{24}} \Delta_{24} + \underbrace{(-c_5)}_{\beta_{34}} \Delta_{34}, \quad (5)$$

where the weights α_i and β_{ij} must satisfy the following unbiasedness condition:

$$\begin{aligned} \alpha_1 - \beta_{12} - \beta_{13} &= \sum_{i=1}^5 c_i = 1, \\ \alpha_2 + \beta_{12} - \beta_{24} &= 0, \\ \alpha_3 + \beta_{13} - \beta_{34} &= 0, \\ \alpha_4 + \beta_{24} + \beta_{34} &= 0. \end{aligned} \quad (6)$$

In §4, we will derive a generalized form of this unbiasedness condition, as shown in Eqn. (16).

Using a combined estimator can further reduce the variance. Using the variance values defined in Table 1, we can compute the variance of the combined estimator H as

$$\begin{aligned} \mathbb{V}(H) &= \sum_{i=1}^4 \alpha_i^2 \mathbb{V}(I_i) + \sum_{(i,j) \in G} \beta_{ij}^2 \mathbb{V}(\Delta_{ij}) \\ &= \alpha_1^2 + 4\alpha_2^2 + 4\alpha_3^2 + 8\alpha_4^2 + 2\beta_{12}^2 + 2\beta_{13}^2 + 2\beta_{24}^2 + 2\beta_{34}^2, \end{aligned} \quad (7)$$

where $G = \{(1, 2), (1, 3), (2, 4), (3, 4)\}$ is the set of neighboring pixel pairs. By minimizing $\mathbb{V}(H)$ subject to the unbiasedness condition expressed in Eqn. (6), we can find the optimal weights:

$$\begin{aligned} \alpha_1 &= 0.7250, \alpha_2 = 0.1125, \alpha_3 = 0.1125, \alpha_4 = 0.0500, \\ \beta_{12} &= -0.1375, \beta_{13} = -0.1375, \beta_{24} = -0.0250, \beta_{34} = -0.0250. \end{aligned} \quad (8)$$

The minimum variance of the reconstruction result is 0.725, which is lower than the original variance $\mathbb{V}(I_1) = 1$. A generalized version of this optimization problem is presented in §5.

Relation to $\mathcal{L}_2$ Poisson Reconstruction. The $\mathcal{L}_2$ Poisson reconstruction is an unbiased approach that is commonly used in gradient-domain rendering [Lehtinen et al. 2013]. It solves the following optimization problem:

$$\min_{\tilde{I}} \sum_{(i,j) \in G} \|\tilde{I}_j - \tilde{I}_i - \Delta_{ij}\|^2 + \lambda \sum_{i=1}^4 \|\tilde{I}_i - I_i\|^2. \quad (9)$$

The reconstructed pixel color $\tilde{I}_1$ can also be expressed in the form of

$$\tilde{I}_1 = \sum_{i=1}^4 \alpha_i I_i + \sum_{(i,j) \in G} \beta_{ij} \Delta_{ij}. \quad (10)$$

Given $\lambda = 0.1$, which is commonly used in practice, we solve this optimization problem and find the weights as

$$\begin{aligned} \alpha_1 &= 0.2799, \alpha_2 = 0.2439, \alpha_3 = 0.2439, \alpha_4 = 0.2323, \\ \beta_{12} &= -0.3600, \beta_{13} = -0.3600, \beta_{24} = -0.1161, \beta_{34} = -0.1161. \end{aligned} \quad (11)$$

It is straightforward to verify that these weights satisfy the unbiasedness condition in Eqn. (6), which is consistent with the fact that $\mathcal{L}_2$ Poisson reconstruction is unbiased; we also demonstrate a proof in Appendix A. The variance of the reconstruction result is approximately 1.5, suggesting that $\mathcal{L}_2$ Poisson reconstruction produces sub-optimal results with larger errors.

4 Unbiasedness Condition of Image Reconstruction

Having shown promising results on a simple 2×2 problem, we now generalize our method to full-size images. We first present the formulation of generalized image reconstruction (§4.1), and then we derive the condition that ensures our reconstruction to be unbiased (§4.2). In the end, we show the matrix form of the unbiasedness condition in the context of gradient-domain rendering (§4.3).

4.1 Generalized Image Reconstruction

Suppose we have an image I , whose resolution is $N = w \times h$, rendered by an unbiased Monte Carlo rendering algorithm such as path tracing, as well as the unbiased estimates of pixel differences Δ_{ij} for any pixel pair $1 \leq i < j \leq N$, our goal is to reconstruct pixel colors using linear combinations of the estimated pixel colors and differences. Specifically, for any pixel p , the reconstructed pixel color

$$\hat{I}_p = \sum_{i=1}^N \alpha_i^{(p)} I_i + \sum_{i=1}^N \sum_{j=i+1}^N \beta_{ij}^{(p)} \Delta_{ij}, \quad (12)$$

where $\alpha_i^{(p)}$ and $\beta_{ij}^{(p)}$ are the weights of linear combination.

Note that our reconstruction formulated in Eqn. (12) is a generalized expression compared with existing image denoising or reconstruction methods. For example, denoising methods such as bilateral filtering [Kopf et al. 2007] and kernel prediction [Kalantari et al. 2015; Bako et al. 2017; Vogels et al. 2018] can be viewed as special cases of our method, where the weights of pixel differences $\beta_{ij}^{(p)}$ are set to zero. In §4.3 and §5, we will show an application of our reconstruction technique to gradient-domain rendering [Lehtinen et al. 2013; Kettunen et al. 2015], where the weights $\beta_{ij}^{(p)}$ are non-zero only for adjacent pixel pairs. Nevertheless, we present the general formulation as shown in Eqn. (12), and we hope this will be useful when the estimators for color differences of arbitrary pixel pairs become more practical.

4.2 Unbiasedness Condition

Given unbiased estimates of pixel colors I_i and their differences Δ_{ij} , we want to ensure that the reconstructed color for any pixel p is also unbiased, i.e., $\mathbb{E}(\hat{I}_p) = \mathbb{E}(I_p)$. Taking expectation of Eqn. (12), we have

$$\mathbb{E}(\hat{I}_p) = \sum_{i=1}^N \mathbb{E}(\alpha_i^{(p)} I_i) + \sum_{i=1}^N \sum_{j=i+1}^N \mathbb{E}(\beta_{ij}^{(p)} \Delta_{ij}). \quad (13)$$

Assume that the weights $\alpha_i^{(p)}, \beta_{ij}^{(p)}$ and the estimated quantities I_i, Δ_{ij} are uncorrelated, i.e.,

$$\mathbb{E}(\alpha_i^{(p)} I_i) = \alpha_i^{(p)} \mathbb{E}(I_i) \text{ and } \mathbb{E}(\beta_{ij}^{(p)} \Delta_{ij}) = \beta_{ij}^{(p)} \mathbb{E}(\Delta_{ij}). \quad (14)$$

By substituting $\mathbb{E}(\Delta_{ij}) = \mathbb{E}(I_j) - \mathbb{E}(I_i)$, the RHS of Eqn. (13) can be rewritten as

$$\mathbb{E}(\hat{I}_p) = \sum_{i=1}^N \left(\alpha_i^{(p)} + \sum_{k=1}^{i-1} \beta_{ki}^{(p)} - \sum_{j=i+1}^N \beta_{ij}^{(p)} \right) \mathbb{E}(I_i). \quad (15)$$

Therefore, the unbiasedness condition for any pixel p , i.e., $\mathbb{E}(\hat{I}_p) = \mathbb{E}(I_p)$, is equivalent to

$$\alpha_i^{(p)} + \sum_{k=1}^{i-1} \beta_{ki}^{(p)} - \sum_{j=i+1}^N \beta_{ij}^{(p)} = \delta_{ip} = \begin{cases} 1, & \text{if } i = p, \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

The key assumption of the unbiasedness condition is that the weights and the Monte Carlo estimates are uncorrelated, as shown in Eqn. (14). In prior work, Back et al. [2023] used Hogg's theorem [Hogg 1960] to design input-dependent uncorrelated kernels. However, the uncorrelated statistics does not hold in general, especially when the sample counts of Monte Carlo estimators are low, leading to biased reconstruction results. Instead, we use the dual-buffer technique (§5.3) to decorrelate the weights and the image estimates. Figures 9 and 10 validate the unbiasedness of our method.

4.3 Matrix Form of Unbiasedness Condition for Gradient-Domain Rendering

Having derived the unbiasedness condition for generalized image reconstruction, we now apply this to gradient-domain rendering. In gradient-domain rendering, we only compute pixel differences (gradients) between neighboring pixels, so we have non-zero weights $\beta_{ij}^{(p)}$ only if pixel i and pixel j are adjacent, which simplifies the unbiasedness condition.

For compactness, we want to rewrite the unbiasedness condition described in Eqn. (16) into a matrix form. Given a pixel p , we arrange its reconstruction weights $\alpha_i^{(p)}$ and $\beta_{ij}^{(p)}$ into a vector $\vec{w}_p$:

$$\vec{w}_p = (\alpha_1^{(p)}, \alpha_2^{(p)}, \dots, \alpha_N^{(p)}, \beta_{12}^{(p)}, \dots, \beta_{ij}^{(p)}, \dots, \beta_{(N-1)N}^{(p)})^\top, \quad (17)$$

where $\beta_{ij}^{(p)}$ denotes the weight of pixel gradients between two neighboring pixels i and j . Assuming the image resolution to be $w \times h$, the weight vector $\vec{w}_p$ has a size of

$$\underbrace{wh}_{\alpha_i^{(p)}} + \underbrace{w(h-1) + (w-1)h}_{\text{number of } \beta_{ij}^{(p)}} = 3wh - w - h = 3N - w - h, \quad (18)$$

rather than a size of $N(N+1)/2$ as in the general formulation. By factoring out the weights and their coefficients ± 1 in Eqn. (16), we have a matrix S with a size of N rows and $(3N - w - h)$ columns, where each row corresponds to a pixel and each column corresponds to a weight entry of $\vec{w}_p$. This matrix S is sparse, and each row has at most 5 non-zero elements (corresponding to the pixel itself and its four neighbors), whose values are either 1 or -1. Ultimately, the matrix form of the unbiasedness condition for pixel p can be written

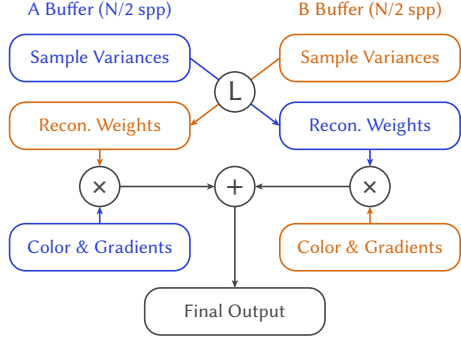


Fig. 2. Flowchart of the dual-buffer technique. The input is split into two buffers. The reconstruction weights derived from the sample variance of buffer A are applied to the color and gradient of buffer B, and vice versa. The final output is the average of two reconstructed images.

as:

$$S\vec{w}_p = \vec{e}_p, \quad \text{where } \vec{e}_p = (0, \dots, 0, \underbrace{1}_{p\text{-th}}, 0, \dots, 0)^\top. \quad (19)$$

By column-wise concatenating the vectors $\vec{w}_p$ of all pixels, i.e., $W = (\vec{w}_1, \dots, \vec{w}_N)$, we can express the unbiasedness condition for the entire image as

$$SW = \mathbf{I}_N, \quad (20)$$

where $\mathbf{I}_N$ is the identity matrix of size N .

5 Variance-Minimizing Reconstruction for Gradient-Domain Rendering

In gradient-domain rendering, image reconstruction is a crucial component. It is desirable to have a robust unbiased reconstruction technique. The unbiasedness condition derived in §4 describes the solution space of unbiased reconstruction. The classical $\mathcal{L}_2$ screened Poisson reconstruction is one of the feasible solutions, but its reconstruction results are often sub-optimal.

In this section, we propose a practical unbiased image reconstruction algorithm by finding the optimal solution in the solution space constrained by the unbiasedness condition. Our goal is to minimize a non-linear objective function, subject to the unbiasedness condition described in Eqn. (20):

$$\min_W \mathcal{L}(\hat{I}(W)), \quad \text{s.t. } SW = \mathbf{I}_N. \quad (21)$$

The optimization objective $\mathcal{L}(\hat{I})$ is a function of the reconstructed image, which depends on the weights W (i.e., $\alpha_i^{(p)}$ and $\beta_{ij}^{(p)}$). There is large design space for the optimization objectives. We introduce a loss function based on the *sample variance* from Monte Carlo rendering (§5.1). Next, we demonstrate how to solve the optimization problem in practice (§5.2).

5.1 Loss Function

One of the commonly used objective functions is the mean squared error (MSE) between the reconstructed image $\hat{I}$ and the ground truth

image I^* :

$$\frac{1}{N} \sum_{p=1}^N (\hat{I}_p - I_p^*)^2. \quad (22)$$

However, in the context of Monte Carlo rendering, the ground truth image is not generally available. Therefore, we choose to minimize the sample variance $S^2(\cdot)$ of the reconstructed image, since variance is the main source of error in unbiased Monte Carlo rendering. This loss function purely depends on the Monte Carlo estimates computed using a finite number of samples.

We express the loss function as the sum of each pixel's sample variance in the reconstructed image:

$$\mathcal{L}(\hat{I}) = \sum_{p=1}^N S^2(\hat{I}_p). \quad (23)$$

By plugging in Eqn. (12) and assuming that pixel colors and gradients are uncorrelated, we have

$$S^2(\hat{I}_p) = \sum_{i=1}^N S^2(\alpha_i^{(p)} I_i) + \sum_{i=1}^N \sum_{j \in \text{adj}(i)} S^2(\beta_{ij}^{(p)} \Delta_{ij}), \quad (24)$$

where $\text{adj}(i) = \{j \mid j > i \text{ and } j \in \text{Neighbors}(i)\}$ is the set of neighboring pixels below and to the right of pixel i (the definition of $\beta_{ij}^{(p)}$ requires $j > i$). Furthermore, we assume that $\alpha_i^{(p)}$ and I_i , as well as $\beta_{ij}^{(p)}$ and Δ_{ij} , are uncorrelated (same as in Eqn. (14) for the unbiasedness condition), which allows us to simplify the sample variance as:

$$S^2(\hat{I}_p) = \sum_{i=1}^N (\alpha_i^{(p)})^2 S^2(I_i) + \sum_{i=1}^N \sum_{j \in \text{adj}(i)} (\beta_{ij}^{(p)})^2 S^2(\Delta_{ij}). \quad (25)$$

Estimating and filtering sample variance. During gradient-domain path tracing, we sample light paths to unbiasedly estimate pixel colors I_i and their gradients Δ_{ij} , and the sample variances $S^2(I_i)$ and $S^2(\Delta_{ij})$ can also be estimated on-the-fly as a byproduct. However, when the sample count per pixel is insufficient, the estimated sample variance may be noisy and less reliable. To mitigate this issue, we apply a joint bilateral filter to smooth the sample variance. Specifically, we set the filter radius to be five pixels. For filtering the sample variance of pixel colors $S^2(I_i)$, we use the albedo and normal G-buffers as auxiliary buffers to guide filtering. The filtered sample variance at pixel i is

$$\hat{S}^2(I_i) = \frac{1}{k_i} \sum_{j \in \Omega_i} S^2(I_j) G_{\sigma_s}(\|i - j\|) G_{\sigma_a}(\|A_i - A_j\|), \quad (26)$$

where $G_\sigma(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right)$ is a Gaussian kernel with standard deviation σ , the set Ω_i contains the neighboring pixels around i within the filter radius, the normalization factor k_i equals to the sum of weights $\sum_{j \in \Omega_i} G_{\sigma_s} G_{\sigma_a}$, and the auxiliary feature vector $A_i = [\vec{n}_i, \vec{\alpha}_i]$ is the concatenation of the normal vector $\vec{n}_i$ and albedo $\vec{\alpha}_i$, normalized by the maximum absolute entry value of this vector. We chose to set σ_a to 0.01 and σ_s to 3.

For filtering the sample variance of pixel gradients $S^2(\Delta_{ij})$, we use a different auxiliary feature vector $A_{ij} = [\vec{n}_i, \vec{n}_j, \vec{\alpha}_i, \vec{\alpha}_j]$, accounting

for the features of both neighboring pixels involved in the gradient. The filtered sample variance of the gradient Δ_{ij} is

$$\hat{S}^2(\Delta_{ij}) = \frac{1}{k_{ij}} \sum_{(p,q) \in \Omega_{ij}} S^2(\Delta_{pq}) G_{\sigma_s}(\|i - p\|) G_{\sigma_a}(\|A_{ij} - A_{pq}\|), \quad (27)$$

where the set Ω_{ij} contains the neighboring pixel pairs that have the same gradient direction as Δ_{ij} , and k_{ij} is the corresponding normalization factor.

Matrix form. Substituting the filtered sample variance into Eqn. (25), we can define the loss function for a single pixel p in a quadratic form:

$$\mathcal{L}(\hat{I}_p) = \sum_{i=1}^N (\alpha_i^{(p)})^2 \hat{S}^2(I_i) + \sum_{i=1}^N \sum_{j \in \text{adj}(i)} (\beta_{ij}^{(p)})^2 \hat{S}^2(\Delta_{ij}) = \vec{w}_p^\top F \vec{w}_p, \quad (28)$$

where F is a diagonal matrix written as

$$F = \text{diag}(\hat{S}^2(I_1), \dots, \hat{S}^2(I_N), \hat{S}^2(\Delta_{12}), \dots, \hat{S}^2(\Delta_{(N-1)N})). \quad (29)$$

Recall that the matrix W is the concatenation of all $\vec{w}_p$, the final loss function that accumulates through all pixels is expressed as

$$\mathcal{L}(\hat{I}) = \sum_{p=1}^N \mathcal{L}(\hat{I}_p) = \text{tr}(W^\top F W), \quad (30)$$

where $\text{tr}(\cdot)$ computes the trace of a matrix.

5.2 Solving Optimization

Our unbiased image reconstruction boils down to solving the following quadratic optimization problem with linear constraints:

$$\min_W \text{tr}(W^\top F W), \quad \text{s.t.} \quad S W = \mathbf{I}_N. \quad (31)$$

We use the method of Lagrange multipliers to solve this constrained optimization problem. The Lagrangian is defined as

$$\mathcal{L}' = \text{tr}(W^\top F W) + \text{tr}(\Lambda(S W - \mathbf{I}_N)), \quad (32)$$

where Λ is the Lagrange multiplier matrix. By setting the derivatives $\partial \mathcal{L}' / \partial W$ and $\partial \mathcal{L}' / \partial \Lambda$ to zero, we get the optimal solution as

$$W = F^{-1} S^\top (S F^{-1} S^\top)^{-1}. \quad (33)$$

Finally, the reconstructed image can be computed as

$$\hat{I} = W^\top \vec{c}, \quad \text{where } \vec{c} = (I_1, \dots, I_N, \Delta_{12}, \Delta_{23}, \dots, \Delta_{(N-1)N})^\top, \quad (34)$$

which is the matrix form of Eqn. (12).

Note that the matrix W has a size of $(3N - w - h) \times N$, which is extremely large and dense. Fortunately, we do not have to explicitly compute W , since our goal is to compute the final reconstructed image

$$\hat{I} = W^\top \vec{c} = (S F^{-1} S^\top)^{-1} S F^{-1} \vec{c}. \quad (35)$$

The final image $\hat{I}$ can be computed by solving the following linear system

$$(S F^{-1} S^\top) \hat{I} = S F^{-1} \vec{c} \quad (36)$$

using the conjugate gradient methods.

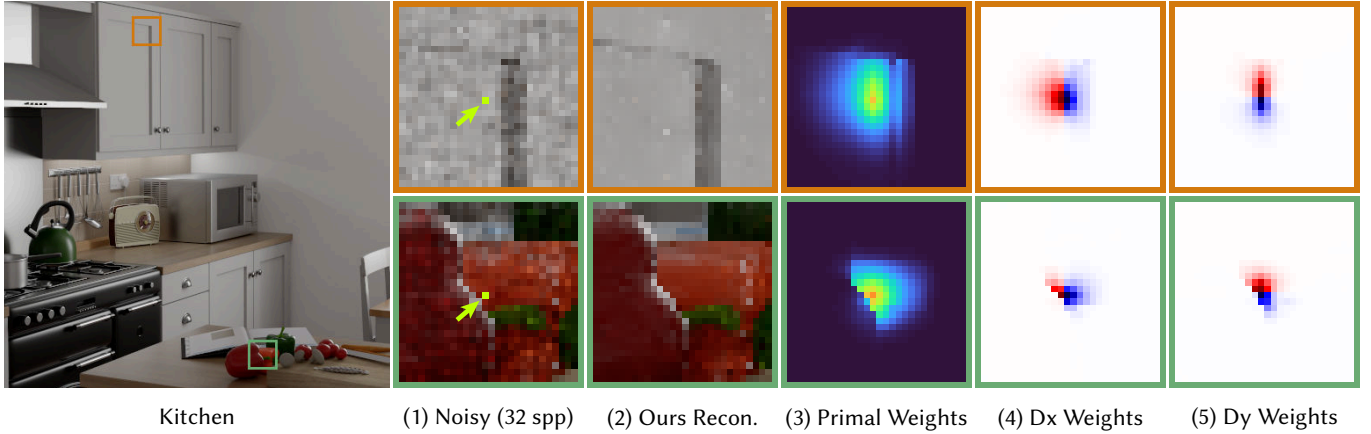


Fig. 3. Visualization of reconstruction weights for the lime-green pixel highlighted in (1). (2) shows close-up views of the reconstructed images around the target pixels. (3) illustrates the spatial distribution of the primal weights α_i for colors. (4) and (5) depict our method's horizontal and vertical weights β_{ij} for gradients, where red indicates positive, blue indicates negative, and white indicates zero. These demonstrate how our method leverages color and gradient information from relevant regions while avoiding noisy or unrelated areas, such as across sharp edges.

5.3 Dual-Buffer Technique

So far, we have made the following uncorrelation assumptions:

- (1) Reconstruction weights and input colors/gradients are uncorrelated (Eqn. (14));
- (2) Pixel colors and pixel gradients are uncorrelated (Eqn. (24));
- (3) Reconstruction weights and the sample variances of input colors/gradients are uncorrelated (Eqn. (25)).

Note that only assumption (1) affects the unbiasedness of reconstruction results; in theory, violating assumption (2) or (3) just affects the optimality of variance minimization. Although one can independently estimate pixel colors I_i and their gradients Δ_{ij} , it significantly increases the rendering time and thus is not practical. In our implementation, we follow the standard gradient-domain path tracing algorithm [Kettunen et al. 2015] and estimate pixel gradients using shift mapping. On the other hand, for the critical uncorrelation assumption (1) formulated in Eqn. (14), we use the following dual-buffer technique to enforce unbiasedness.

To satisfy the uncorrelation assumption in Eqn. (14), we use the dual-buffer technique introduced by Rousselle et al. [2012] as illustrated in Fig. 2. The Monte Carlo rendering samples are divided into two halves. The reconstruction weights of the first half are computed using the estimated sample variance of the second half and vice versa. This technique guarantees independence between the weights and the input estimates, at the cost of using noisier estimates since the effective sample count is halved. We also implemented a biased version using only a single buffer. More details and comparisons are discussed in §6.1.3.

6 Experiments

We use gradient-domain path tracing [Kettunen et al. 2015] implemented in LuisaRender [Zheng et al. 2022] to generate input data, including pixel colors, pixel gradients, auxiliary buffers, and per-pixel sample variances. All scenes are from Rendering Resources

[Bitterli 2016]. Our unbiased reconstruction, which solves the linear system in Eqn. (36), is implemented using a conjugate gradient solver from the GPU sparse matrix library CuSparse. We set the solver's tolerance to 10^{-6} and the maximum iteration count to 50K. To ensure the optimization problem is well-posed, we clamp the diagonal values of matrix F (see Eqn. (29)) to a minimum of 10^{-4} . We run all the experiments on a workstation equipped with an Intel i9-12900KF CPU and an Nvidia 3090 GPU. The default image resolution is 1280×720 .

6.1 Analysis and Evaluation

We now analyze several components of our method and evaluate different variants by ablation studies to validate our design choices.

6.1.1 Visualization of reconstruction weights. Fig. 3 visualizes the reconstruction weights computed by our algorithm for a representative pixel. The spatial distribution of weights demonstrates our method's ability to adaptively select relevant information. For instance, the primal weights for pixel colors tend to borrow from similarly colored local regions, while the Dx and Dy weights for gradients draw from areas that indicate local structures, crucially avoiding crossing sharp edges or incorporating information from noisy or irrelevant samples. This illustrates how our method effectively leverages both primal and gradient data, which is key to minimizing variance while preserving detail.

6.1.2 Filtering sample variance. Fig. 4 evaluates different choices of kernel parameters (i.e., diameter d , spatial sigma σ_a , and feature sigma σ_s) for filtering sample variance. Directly using sample variances without filtering may produce unsatisfactory results, especially in scenes with low sample rates, because the noise in sample variance could misguide the reconstruction algorithm. In contrast, using filtered variance leads to lower reconstruction errors. We find that the quality of the reconstructed results is not quite sensitive to choices of kernel parameters. In our experiments, we use $(d, \sigma_a, \sigma_s) = (11, 0.01, 3)$ by default.

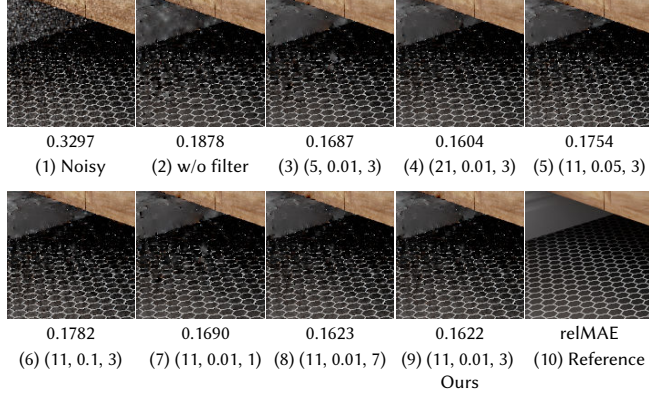


Fig. 4. Comparisons of sample variance filtering on the Bathroom scene at 16 samples per pixel. (1) is the noisy input image, and (2) show the constructed image using sample variance without filtering. From (3)–(9), we perform an ablation study on different choices of filter kernel parameters. By default, we choose the diameter of the filter kernel, the spatial sigma, and the feature sigma to be $(d, \sigma_d, \sigma_s) = (11, 0.01, 3)$.



Fig. 5. Comparisons between directly using sample variances (Single w/o filter), using filtered sample variances (Single w/ filter), and using modified sample variances (Single w/ mod. filter) for our single-buffer variant.

6.1.3 Single-buffer slightly biased variant. Our primary method employs a dual-buffer strategy to ensure strict unbiasedness. Here, we also investigate a single-buffer variant where input samples are not split. In this variant, the sample variances used for weight computation are correlated with pixel colors and gradients, which violates the unbiasedness condition in Eqn. (14) but can offer practical advantages. At the cost of becoming slightly biased, this single-buffer approach can better react to outliers because the input colors and gradients use effectively $2\times$ more samples, possibly leading to more accurate estimates of sample variances and reconstruction weights. Moreover, the presence of correlation allows for modifying the sample variance to take into account the input colors and gradients, thereby enhancing the algorithm’s ability to detect and adapt to outliers.



Fig. 6. Ablation study on the trade-off between color and gradient spp at equal rendering time. Extra color spp are obtained by replacing G-PT with PT, which reduces gradient spp and leads to degraded reconstruction quality.

For the single-buffer variant, we find that directly using sample variances $S^2(\cdot)$ as the estimators of variances would lead to darkened outputs, as darker places tend to result in low sample variances, misleading our algorithm to preserve these values. On the other hand, using the filtered sample variance $\hat{S}^2(\cdot)$, as in the dual-buffer version, loses sensitivity to outliers, making it difficult to eliminate them. As a trade-off between visual quality and bias, we design the following modified sample variance $S_{\text{mod}}^2(\cdot)$, replacing the filtered sample variance $\hat{S}^2(\cdot)$ used in the diagonal matrix F (Eqn. (29)). The modified sample variance is based on the filtered sample variance and compensates for both darker and brighter regions:

$$S_{\text{mod}}^2(x) = \hat{S}^2(x) + |\hat{S}^2(x) - S^2(x)|. \quad (37)$$

In Fig. 5, we evaluate our choices among $S^2(\cdot)$, $\hat{S}^2(\cdot)$, and $S_{\text{mod}}^2(\cdot)$. The results verify our previous observations: directly using the sample variance produces smooth but overly dark results; using the filtered sample variance yields numerically good values but slightly worse visual quality; the modified sample variance achieves the best overall performance. Therefore, we choose to adopt the modified sample variance for robustness in this single-buffer variant.

6.1.4 Balancing Color and Gradient Sample Counts. In Fig. 6, we investigate the effect of varying the allocation between color and gradient samples under equal rendering time. Specifically, we reduce the number of gradient samples and compensate by increasing color samples by replacing each gradient path tracing (G-PT) sample with two standard path tracing (PT) samples, leveraging the fact that PT runs twice as fast as G-PT in our setup. The total color spp equals the sum of G-PT and PT samples, while the gradient spp comes solely from G-PT. We observe that fewer gradient samples diminish the benefits of G-PT’s low-variance gradient estimation, resulting in lower reconstruction quality. The best results are achieved when all samples are allocated to G-PT.

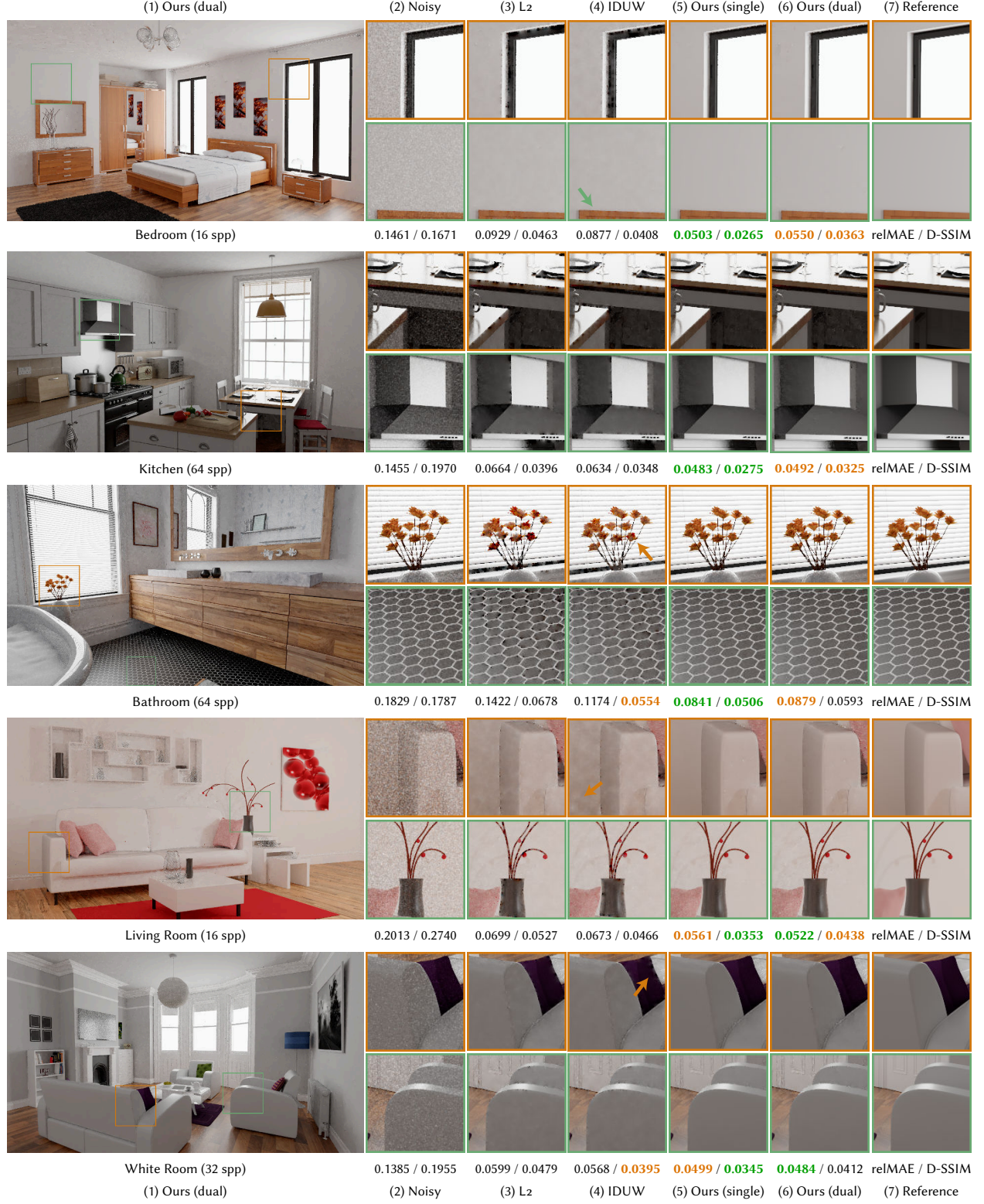


Fig. 7. Reconstruction results comparing to existing unbiased (or slightly biased) approaches: L_2 Poisson reconstruction, *Input-dependent uncorrelated weighting for Monte Carlo denoising* (IDUW) [Back et al. 2023], and our single-buffer variant. Relative MAE(relMAE) and D-SSIM are used as metrics. The **best** and **second-best** values are highlighted. Results show that our method could produce better outputs without violating the unbiasedness property.

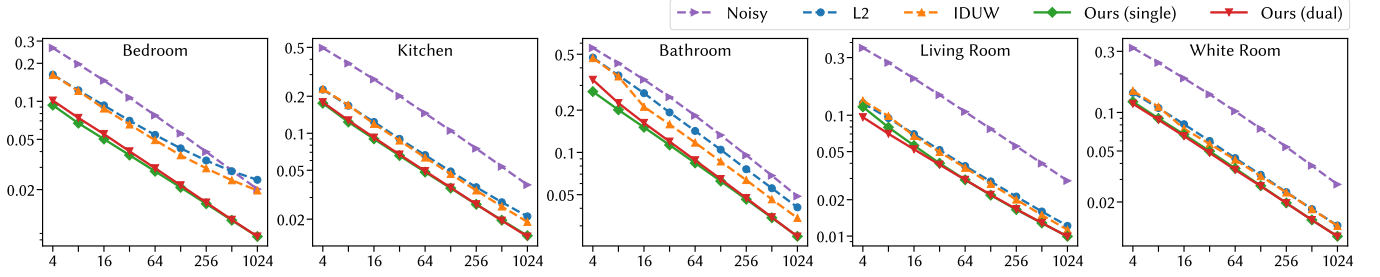


Fig. 8. Error curves of relMAE with respect to the number of samples per pixel.

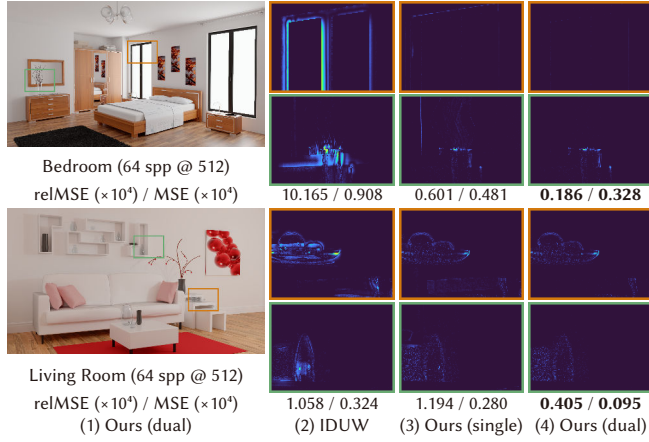


Fig. 9. Bias comparison among IDUW [Back et al. 2023], the single-buffer variant of our method, and our primary method using the dual-buffer technique. Images are reconstructed and averaged from 512 iterations, and each iteration uses 64 spp. The image error is dominated by bias when a high iteration count is used. (2)–(4) show error maps in false color, where brighter pixels represent larger error values. Our method has the lowest bias, and our single-buffer variant also shows significantly less bias than IDUW.

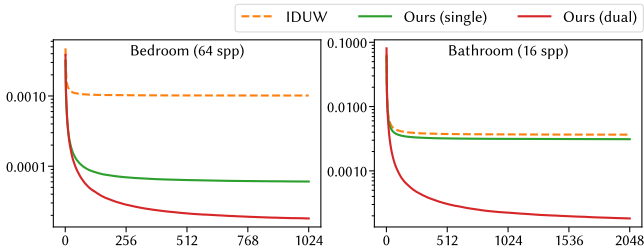


Fig. 10. Plots of relMSE curves with an increasing number of iterations. As the iteration number increases, bias gradually dominates the image error. Among the three methods, ours using the dual-buffer technique exhibits the lowest bias, which approaches zero closely. The single-buffer variant of our method is slightly biased, but it still outperforms the other slightly biased baseline IDUW.

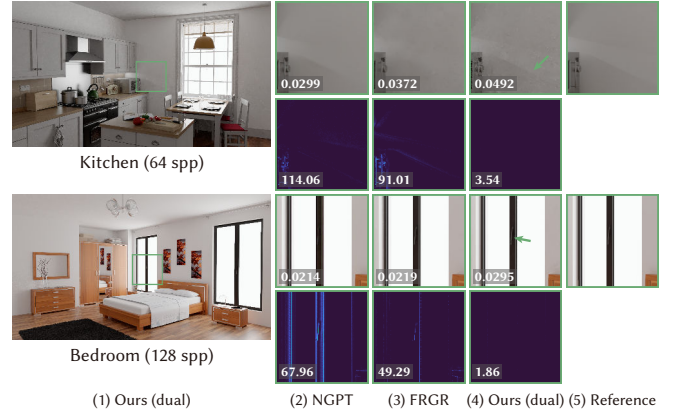


Fig. 11. Comparisons between our method, NGPT [Kettunen et al. 2019] and FRGR [Yan et al. 2024]. The metric used to compare reconstruction results (the first row of close-up images) is relMAE. The metric for comparing bias (the second row of close-up images) is relMSE scaled by 10^5 . The bias map shows the image error between the ground truth and the averaged image from 512 iterations of rendering. The image error is dominated by bias when a high iteration count is used.

6.2 Comparisons

In this section, we compare our method to two unbiased (or slightly biased) reconstruction methods, i.e., the $\mathcal{L}_2$ Poisson reconstruction, and IDUW [Back et al. 2023]. We also compare our approach with other biased reconstruction approaches, including FRGR [Yan et al. 2024], NGPT [Kettunen et al. 2019], AFGSA [Yu et al. 2021], and NPPD [Balint et al. 2023]. The metrics we used for comparison include relative mean absolute error (relMAE) and data structural similarity index (D-SSIM). In both metrics, a smaller value indicates a reconstruction closer to the ground truth.

6.2.1 Comparisons to unbiased reconstruction methods. We compared our method to the unbiased baseline $\mathcal{L}_2$ Poisson reconstruction and the slightly biased method IDUW [Back et al. 2023]. In Fig. 7, both the dual-buffer version of our method and the single-buffer variant outperform other methods in terms of visual quality and numeric metrics. Notably, compared to the baselines, our method generates significantly fewer artifacts in clean regions and regions near edge boundaries.

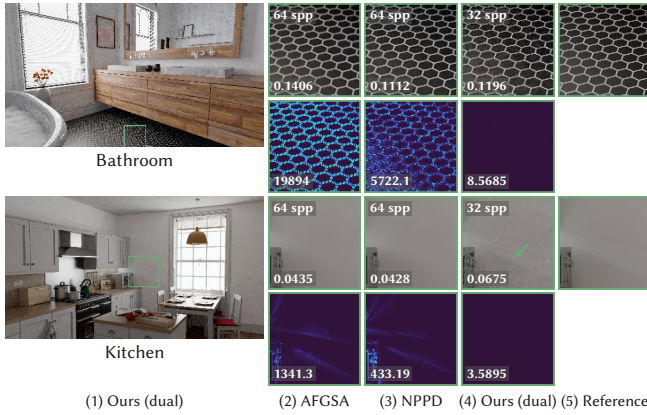


Fig. 12. Comparisons between our method, AFGSA [Yu et al. 2021] and NPPD [Balint et al. 2023]. The metric used to compare reconstruction results (the first row of close-up images) is relMAE. The metric for comparing bias (the second row of close-up images) is relMSE scaled by 10^5 . The bias map shows the image error between the ground truth and the averaged image from 512 iterations of rendering. The image error is dominated by bias when a high iteration count is used.

Fig. 8 shows the error curves of all methods with respect to sampling rates, where our method shows consistently better results than other methods. Notably, the single-buffer variant of our method shows almost the same performance as the dual-buffer version.

Bias evaluation. We design the following experiment to verify unbiasedness. At each iteration, we run gradient-domain path tracing with a relatively low sample count (e.g., 16 spp) and reconstruct images by our method (using dual-buffer), our single-buffer variant, and IDUW [Back et al. 2023], respectively. We then obtain a final image by averaging the reconstructed images from multiple iterations. When the iteration increases, the image variance approaches zero, and thus the error is dominated by bias. Fig. 9 shows visualizations of error maps from images computed using 512 iterations with 64 spp per iteration. Fig. 10 shows plots of error curves with increasing iterations. These results indicate that our single-buffer variant produces a larger bias compared to the unbiased dual-buffer one, but its bias is still significantly smaller than IDUW [Back et al. 2023].

Performance. An unoptimized implementation of our method takes around 10 seconds to reconstruct a 1280×720 image. In contrast, the time for a 1280×720 image by $\mathcal{L}_2$ Poisson reconstruction is around 1 second, and 0.5 second by IDUW [Back et al. 2023]. Although our reconstruction takes longer time, it is usually much shorter than the rendering time and hence it is not the bottleneck.

6.2.2 Comparisons to biased reconstruction methods. In Fig. 11 and Fig. 12, we also compare our method to biased reconstruction methods, including both gradient-domain methods (FRGR [Yan et al. 2024] and NGPT [Kettunen et al. 2019]) and non-gradient-domain methods (AFGSA [Yu et al. 2021] and NPPD [Balint et al. 2023]). For the comparisons to AFGSA and NPPD, we provide AFGSA and NPPD with twice the sample rate to match the rendering time as

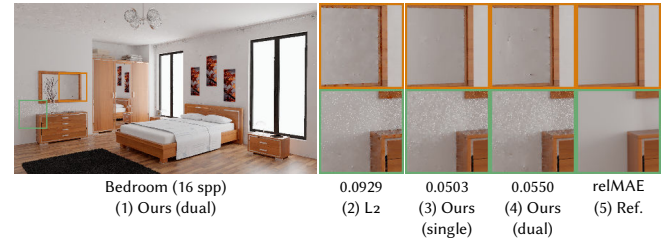


Fig. 13. A visualization of the outliers in our method compared with $\mathcal{L}_2$ Poisson reconstruction. The first row shows dipole-like artifacts caused by gradient errors, which our dual-buffer method struggles to remove. The second row shows typical outliers resulting from errors in the color space.

they do not need gradients, and crop the inputs from 1280×720 to 1280×706 , to satisfy the requirements of NPPD. We show reconstructions and bias maps in close-up views for all methods. In both figures, the first row of close-up views shows reconstructed images and the second row shows bias maps.

From the results, we could find that those biased approaches introduce much larger biases. In contrast, our method produces slightly larger reconstruction errors to ensure strict unbiasedness. Nevertheless, our method is still capable of producing comparable results and can preserve shadow edges while avoiding excessive blurring.

7 Conclusion

In this paper, we proposed a novel unbiased reconstruction method for gradient-domain rendering. We introduced a generalized form of linear-based reconstruction that computes pixel values through a linear combination of colors and gradients, and derived a necessary and sufficient condition for unbiasedness as the theoretical foundation. Guided by sample variance, we developed a practical reconstruction algorithm that guarantees unbiasedness while minimizing the variance of reconstruction, achieving superior quality compared to existing unbiased and slightly biased reconstruction methods.

7.1 Limitations

Outliers. While our method guarantees unbiasedness, this property sometimes comes at the cost of visual quality, especially when compared to biased denoising techniques. The most significant issue lies in the presence of outliers. Because our dual-buffer approach determines reconstruction weights using the sample variance from the opposite buffer, the weights are unaware of outliers in the color and gradient values to which they are applied. As a result, more outliers appear in the final output compared to traditional filtering techniques.

Fig. 13 shows examples of outliers produced by our method. These can be roughly divided into two categories. The first are dipole-like artifacts caused by large gradient errors, which our dual-buffer method can hardly remove due to its inability to recognize extreme gradient data. The second are typical outliers in the color space, where both our method and $\mathcal{L}_2$ Poisson reconstruction fail to correct them.

Table 2. Time consumption of our method on different image size

Block size	Ours (dual) (ms)	Ours (single) (ms)
1280×720	13126	6549
512×512	3241	1720
256×256	1012	401
128×128	145	86
64×64	59	47
32×32	47	39

For gradient-space errors, improved sampling strategies could substantially reduce artifacts by producing more accurate gradient estimates. For color-space errors, additional post-processing techniques may be required to mitigate these outliers and achieve more stable results.

A method that can simultaneously enforce unbiasedness through statistical independence while also handling outliers in a robust manner would provide significant improvements in reconstruction quality. However, achieving both goals at once remains challenging, as unbiasedness often limits the ability to adaptively suppress extreme values.

Performance. As mentioned in Sec. 6.2.1, our method requires about 10 seconds for a 1280×720 image. More detailed performance statistics are provided in Table 2. The computation time depends only on the image resolution, so we evaluate on a single scene. Our single-buffer variant, which solves the equation only once, is consistently twice as fast as the dual-buffer algorithm. The solving time is proportional to the number of pixels.

It is worth mentioning that block solving can provide an almost lossless speed-up. In Fig. 3, most weights are concentrated in a small neighborhood around the central pixel, since involving distant pixels introduces large variance, and their weights are therefore automatically reduced to 0. To accelerate computation, one could parallelize block solving with 128×128 patches.

7.2 Future work

We have several potential directions to explore. Our method could be extended to other inputs with gradient estimates or correlated samples, including correlated sampling techniques such as ReSTIR [Bitterli et al. 2020]. Also, our variance-minimizing optimization formulation could be extended to accept arbitrary pixel pairs instead of neighboring pixels as input. Another possible improvement is to accelerate the solving process. In our experiments, we noticed that although the entire image is provided, the reconstruction weights of a certain pixel are always distributed in a local region. If the input image could be semantically segmented, solving a much smaller optimization problem while producing the same solution would be possible. In addition, the choice of loss functions could be further explored.

Acknowledgments

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A Relation to $\mathcal{L}_2$ Poisson Reconstruction

We now prove that the result of $\mathcal{L}_2$ Poisson reconstruction for gradient-domain rendering satisfies our unbiasedness condition expressed in Eqn. (19), indicating $\mathcal{L}_2$ Poisson reconstruction is one of the feasible solutions in the space of unbiased reconstruction methods.

Recall that the $\mathcal{L}_2$ Poisson reconstruction solves the following optimization problem:

$$\min_{\tilde{I}} \sum_{(i,j) \in G} \|\tilde{I}_j - \tilde{I}_i - \Delta_{ij}\|^2 + \lambda \sum_{i=1}^N \|\tilde{I}_i - I_i\|^2, \quad (38)$$

where I_i is the input color of pixel i , Δ_{ij} is the input gradient between pixel j and pixel i , N is the count of pixels, G is the set of neighboring pixel pairs, and $\tilde{I}$ is the reconstructed image. To solve this optimization problem, we need to take the derivative of Eqn. (38) with respect to u and set it to zero. By doing so, we obtain the following linear equation:

$$(L + \lambda I)\tilde{I} = b + \lambda I, \quad (39)$$

where I is an identity matrix. The discrete Laplacian matrix $L \in \mathbb{R}^{N \times N}$ arises from the discretization of the Laplacian operator. For pixel i , assuming it has x neighboring pixels, the entries of L are defined as:

$$L_{ik} = \begin{cases} x, & \text{if } i = k, \\ -1, & \text{if } (i, k) \in G, \\ 0, & \text{otherwise.} \end{cases} \quad (40)$$

The vector $b \in \mathbb{R}^N$ is the divergence of the target gradient field, which is derived from the input gradients Δ_{ij} . For each pixel i , the element b_i is computed as:

$$b_i = \sum_{(i,j) \in G} \Delta_{ij}. \quad (41)$$

The matrix $L + \lambda I$ is symmetric positive definite, and therefore invertible. Let M denote the inverse of $(L + \lambda I)$, i.e., $M = (L + \lambda I)^{-1}$. The result of $\mathcal{L}_2$ Poisson reconstruction can be expressed as:

$$\tilde{I} = \lambda MI + Mb. \quad (42)$$

Expression for a single pixel. Consider a specific pixel p in the image. From Eqn. (42), we can write:

$$\begin{aligned} \tilde{I}_p &= (\lambda MI + Mb)_p \\ &= \lambda \sum_{i=1}^N M_{pi} I_i + \sum_{i=1}^N M_{pi} b_i. \end{aligned} \quad (43)$$

Since b_i is the divergence at pixel i , which equals the sum of gradients over the neighbors of pixel i . Substitute b_i into the Eqn. (43):

$$\begin{aligned} \sum_{i=1}^N M_{pi} b_i &= \sum_{i=1}^N M_{pi} \left(\sum_{(i,j) \in G} \Delta_{ij} \right) \\ &= \sum_{i=1}^N \sum_{(i,j) \in G} M_{pi} \Delta_{ij}. \end{aligned} \quad (44)$$

For each pair of neighboring pixels (i, j) , the gradient term Δ_{ij} appears both in b_i (with coefficient -1 , since $\Delta_{ij} = -\Delta_{ji}$) and b_j

(with coefficient 1). Therefore, the coefficient of Δ_{ij} is $M_{pj} - M_{pi}$ for any $j > i$, so we can express Eqn. (43) as:

$$\tilde{I}_p = \sum_{i=1}^N (\lambda M_{pi}) I_i + \sum_{i=1}^N \sum_{j=i+1}^N (M_{pj} - M_{pi}) \Delta_{ij}. \quad (45)$$

Note that the form of Eqn. (45) matches the form of unbiased reconstruction method described in Eqn. (12). Thus, we can directly identify the weights $\alpha_i^{(p)} = \lambda M_{pi}$ and $\beta_{ij}^{(p)} = M_{pj} - M_{pi}$.

Verification of unbiasedness condition. We can now substitute the weights $\alpha_i^{(p)} = \lambda M_{pi}$ and $\beta_{ij}^{(p)} = M_{pj} - M_{pi}$ into the unbiased condition described in Eqn. (16) and demonstrate that it holds for all pixels p and i . Substituting these weights into Eqn. (16) yields:

$$\begin{aligned} &\alpha_i^{(p)} + \sum_{k=1}^{i-1} \beta_{ki}^{(p)} - \sum_{j=i+1}^N \beta_{ij}^{(p)} \\ &= \lambda M_{pi} + \sum_{k=1}^{i-1} (M_{pi} - M_{pk}) - \sum_{j=i+1}^N (M_{pj} - M_{pi}) \\ &= (\lambda + x)M_{pi} - \sum_{(i,j) \in G} M_{pj}. \end{aligned} \quad (46)$$

Recall that $M = (L + \lambda I)^{-1}$, so we have $M(L + \lambda I) = I$. Using the fact that $L + \lambda I$ is a symmetric matrix, consider the (p, i) -th entry of this matrix:

$$\begin{aligned} I_{pi} &= \sum_{k=1}^N M_{pk} (L + \lambda I)_{ki} \\ &= \sum_{k=1}^N M_{pk} (L + \lambda I)_{ik} \\ &= \lambda M_{pi} + \sum_{k=1}^N M_{pk} L_{ik} \\ &= (\lambda + x)M_{pi} - \sum_{(i,j) \in G} M_{pj} = \delta_{pi} = \delta_{ip}. \end{aligned} \quad (47)$$

We have verified that Eqn. (16) holds. Therefore, the weights solved by $\mathcal{L}_2$ Poisson reconstruction satisfy our proposed unbiased condition.